

Turbocharger



Principles of Operation

The turbocharger is an exhaust-driven centrifugal air compressor. Its purpose is to increase power output by supplying compressed air to the engine.

Expanding exhaust gases drive the turbine shaft assembly to speeds up to 200,000 rpm. Filtered air entering the compressor side of the turbocharger is compressed and delivered through a charge air cooler. The very hot compressed air is cooled by the charge air cooler, then continues on to fill the intake manifold at a higher pressure than atmospheric pressure. Because considerably more air is forced into the intake manifold, the results are increased power, fuel efficiency and the ability to maintain power at higher altitudes.

The turbocharger is governed by a electric wastegate actuator. The electric wastegate actuator redirects some of the exhaust gas past the turbine and therefore acts as a governor. The electric wastegate actuator is controlled by the powertrain control module.

The intake air system is fitted with a turbocharger bypass valve. This vents the intake air system when the throttle is closed and recirculates the air back into the intake air system. It also releases pressure on the turbocharger which will reduce turbo lag and improve acceleration and increase engine performance.

The internal components are oil, coolant and air cooled. Engine oil and coolant are circulated through the center housing which acts as a heat barrier between the "hot" turbine and the "cold" compressor. Bearings are sleeve type and lubricated by engine oil. Oil is circulated to the turbocharger center housing and returned to the engine oil sump through an oil drain in the center housing.

Inspection and Verification

NOTE: This section provides mechanical diagnosis of the turbocharger assembly. If there is a Malfunction Indicator Lamp (MIL) illuminated, or DTCs are present, these should be diagnosed prior to performing turbocharger mechanical diagnosis. Refer to the Powertrain Control/Emissions Diagnosis (PC/ED) manual.

1. Verify the customer concern.
2. Inspect the entire turbocharger system for obvious signs of damage or other mechanical concerns, using the following chart.

Visual Inspection Chart

Mechanical
• Air cleaner element • Air cleaner intake pipe • Air cleaner outlet pipe • Charge air cooler • Charge air cooler intake pipe • Charge air cooler outlet pipe • Coolant leaks • Oil leaks • Turbocharger coolant tubes

Mechanical

- Turbocharger housings
- Turbocharger intake tubes
- Turbocharger oil tubes
- Turbocharger wastegate control actuator linkages
- Wastegate valves

3. If the concern(s) remains after the inspection, GO to [Symptom Chart](#).

4. If a driveability condition still exists, refer to the Powertrain Control/Emissions Diagnosis (PC/ED) manual.

Symptom Chart

Symptom Chart

Condition	Possible Causes	Action
• Loss of power/performance	• Clogged Air Cleaner (ACL) element • Loose connections or damage to air intake hoses and tubes • Malfunctioning fuel system • Poor fuel quality • Engine wear (piston rings, valve guides) • Turbocharger turbine or compressor wheel damage • Damaged exhaust or exhaust leaks at turbocharger housing or exhaust manifolds • Malfunctioning turbocharger bypass valve • Obstructed wastegate actuator linkage	• INSTALL a new ACL element. REFER to Section 303-12. TEST the system for normal operation. • TIGHTEN the hose clamps. INSPECT for damage and REPAIR as necessary. • REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual. • DRAIN and FILL with fresh fuel. • REPAIR as required. • INSPECT the turbocharger. REFER to the Check for Free Rotation — On Vehicle in this section. • INSPECT for leaks. Leaks can usually be detected audibly or visually, by a discoloration caused by escaping hot exhaust gases. REPAIR as required. • REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual. • INSPECT the suspect turbocharger wastegate linkage for obstruction. REPAIR as necessary.

	• Damaged wastegate actuator linkage • Wastegate actuator operation • PCM — DTCs	• INSPECT the suspect turbocharger wastegate linkage for damage. INSTALL a new turbocharger. • TEST the wastegate actuator operation. REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual. • REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual.
• Excessive oil consumption	• Incorrect type or grade of oil • Oil in exhaust manifolds • Clogged ACL element • Blocked or restricted turbocharger oil drain pipes • Damaged or collapsed air intake hoses and tubes • Damaged exhaust or exhaust leaks at turbocharger housing or exhaust manifolds • Turbocharger oil seals leaking • Damaged crankcase ventilation system	• DRAIN and FILL with specified oil. • REPAIR as required. • INSTALL a new ACL element. REFER to Section 303-12. TEST the system for normal operation. • INSPECT the turbocharger oil drain pipes. REPAIR as necessary. • TIGHTEN hose clamps. INSPECT for damage and REPAIR as necessary. • INSPECT for leaks. Leaks can usually be detected audibly or visually, by a discoloration caused by escaping hot exhaust gases. REPAIR as required. • REFER to Turbocharger Internal Oil Leak Test in this section. • Visually INSPECT the crankcase ventilation system.
• Excessive noise	• Turbocharger compressor air intake pipe leaking • Loose connections or damage to air intake hoses and tubes • Air leaks at turbine housing, blown joints or damaged exhaust	• REPAIR or INSTALL a new pipe as required. • TIGHTEN hose clamps. INSPECT for damage and REPAIR as necessary. • INSPECT for leaks. Leaks can usually be detected audibly or visually, by a discoloration caused by escaping hot exhaust gases. REPAIR as required.

	- Carbon build up in the turbine housing causing contact with turbine wheel - Turbocharger imbalance due to foreign object/damage - Change to the wastegate nut location - Inoperative turbocharger bypass valve	- INSPECT the turbocharger. REFER to the Check for Free Rotation — On Vehicle in this section. - REPAIR as required. - INSPECT the turbocharger paint mark location. If the paint marks are not aligned, INSTALL a new turbocharger. - REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual.
Engine emits excessive smoke (black/blue/white)	- Clogged ACL element - Incorrect type or grade of oil - Blocked or restricted turbocharger oil drain pipes - Damaged/restricted or leaking turbocharger compressor air intake pipe - Engine wear (piston rings, valve guides) - Plugged crankcase ventilation system - Turbocharger oil seals leaking	- INSTALL a new ACL element. REFER to Section 303-12. TEST the system for normal operation. - DRAIN and FILL with specified oil. - INSPECT the turbocharger oil drain pipes. REPAIR as necessary. - REPAIR or INSTALL a new pipe as required. - REPAIR as required. - Visually INSPECT the crankcase ventilation system. - REFER to Turbocharger Internal Oil Leak Test in this section.

Condition	Possible Causes	Action
NOTE: The turbocharger oil supply tube is located on the top of the turbocharger. Many leaks from the turbocharger oil supply tube will look like they are turbocharger oil leaks, as the oil will accumulate in the v-band and drip off the bottom of the turbocharger. After replacing the turbocharger oil supply tube, clean the outside of the turbocharger, to avoid false diagnosis of a leaking turbocharger.		
External turbocharger oil leaks	Turbocharger oil tube seals and/or oil tube leaking	- Visually INSPECT the turbocharger oil tubes and the turbocharger oil tube seals/gaskets. - REPLACE the turbocharger oil tube seals/gaskets and the turbocharger oil tubes as needed.
Condition	Possible Causes	Action

• External turbocharger coolant leaks	• Turbocharger coolant tube seals and/or coolant tube leaking	• Visually INSPECT the turbocharger coolant tubes and the turbocharger coolant tube seals/gaskets. • REPLACE the turbocharger coolant tube seals/gaskets and the turbocharger coolant tubes as needed.
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Component Tests

Turbocharger Internal Oil Leak Test

NOTE: *It is normal for a small amount of combustion gas to pass into the crankcase. This gas is scavenged into the air intake system through the positive crankcase ventilation system, which incorporates an crankcase vent oil separator that is part of the valve cover. Some engine oil, in the form of a vapor is carried into the air intake system with the blow-by gases (this engine oil also contributes to valve seat durability). This means that oil will collect inside the air intake components and the turbocharger. This is not an indication that the turbocharger oil seal has failed. The turbocharger oil seal will generally not fail unless the bearings fail first, which will cause the turbocharger to become noisy or seize. Do not install a new turbocharger due to oil inside the turbocharger or the air intake components. If a leak is detected in the oil supply or return tubes or connections, locate and rectify the source. Do not install a new turbocharger due to an oil leak.*

Check the turbocharger compressor inlet for evidence of oil. If excessive oil is present, this indicates that the failure could be in the engine or turbocharger. Refer to Section 303-00 or the following turbocharger check. If excessive oil is found in the turbocharger compressor outlet, check the Charge Air Cooler (CAC) for oil contamination. If contamination is present, flush the CAC. Refer to Section 303-12.

Check the turbocharger turbine outlet for evidence of oil. If excess oil is present in the outlet, remove the turbocharger from the engine and examine the oil supply and return passages in the turbocharger and engine block for restriction.

Check for Free Rotation — On Vehicle

NOTE: *Pre-oil the turbocharger before carrying out this check.*

NOTE: *Turbine and compressor wheels must spin freely when turned by hand. No housing contact is permitted.*

Inspect the turbocharger compressor and turbine blades for damage. If the compressor or turbine wheel blades are damaged, replace the turbocharger.

Press and rotate the turbocharger shaft, then repeat from the opposite side. If either the compressor or the turbine wheel blades contact the housing, the bearings are worn or damaged and a new turbocharger must be installed.